1) #include <stdio.h>

```
#define FIXED_CHARGE 5000L
#define SLAB1 100
#define SLAB2 200
#define RATE1 0
#define RATE2 225
#define RATE3 450

typedef unsigned short meter_t;

int main(void)
{
   int house_number;
   char category;
   long previous, current;

   unsigned int units;
   long energy_charge = 0;
   long fixed_charge = FIXED_CHARGE;
   long total;

   printf("Enter household number: ");
   if (scanf("%d", &house_number) != 1 || house_number <= 0)
   {
      printf("Invalid household number\n");
      return 1;
   }
```

```
printf("Enter category (D for Domestic, C for Commercial): ");
if (scanf(" %c", &category) != 1)
{
   printf("Invalid category\n");
   return 1;
}

printf("Enter previous meter reading: ");
if (scanf("%ld", &previous) != 1 ||
   previous < 0 || previous > 65535)
{
   printf("Invalid previous reading\n");
   return 1;
}

printf("Enter current meter reading: ");
if (scanf("%ld", &current) != 1 ||
   current < 0 || current > 65535)
{
   printf("Invalid current reading\n");
   return 1;
}

if (current < previous &&
   !(previous >= 65000 && current <= 535))
{
   printf("Invalid meter reading\n");
```

```
    return 1;
}

units = (meter_t)((meter_t)current - (meter_t)previous);

if (units > SLAB2)
{
    energy_charge =
        (SLAB2 - SLAB1) * RATE2 +
        (units - SLAB2) * RATE3;
}
else if (units > SLAB1)
{
    energy_charge = (units - SLAB1) * RATE2;
}
else
{
    energy_charge = units * RATE1;
}

total = energy_charge + fixed_charge;

printf("\n------ ELECTRICITY BILL ------\n");
printf("Household Number: %d\n", house_number);
printf("Category: %c\n", category);
printf("Units Consumed: %u\n", units);

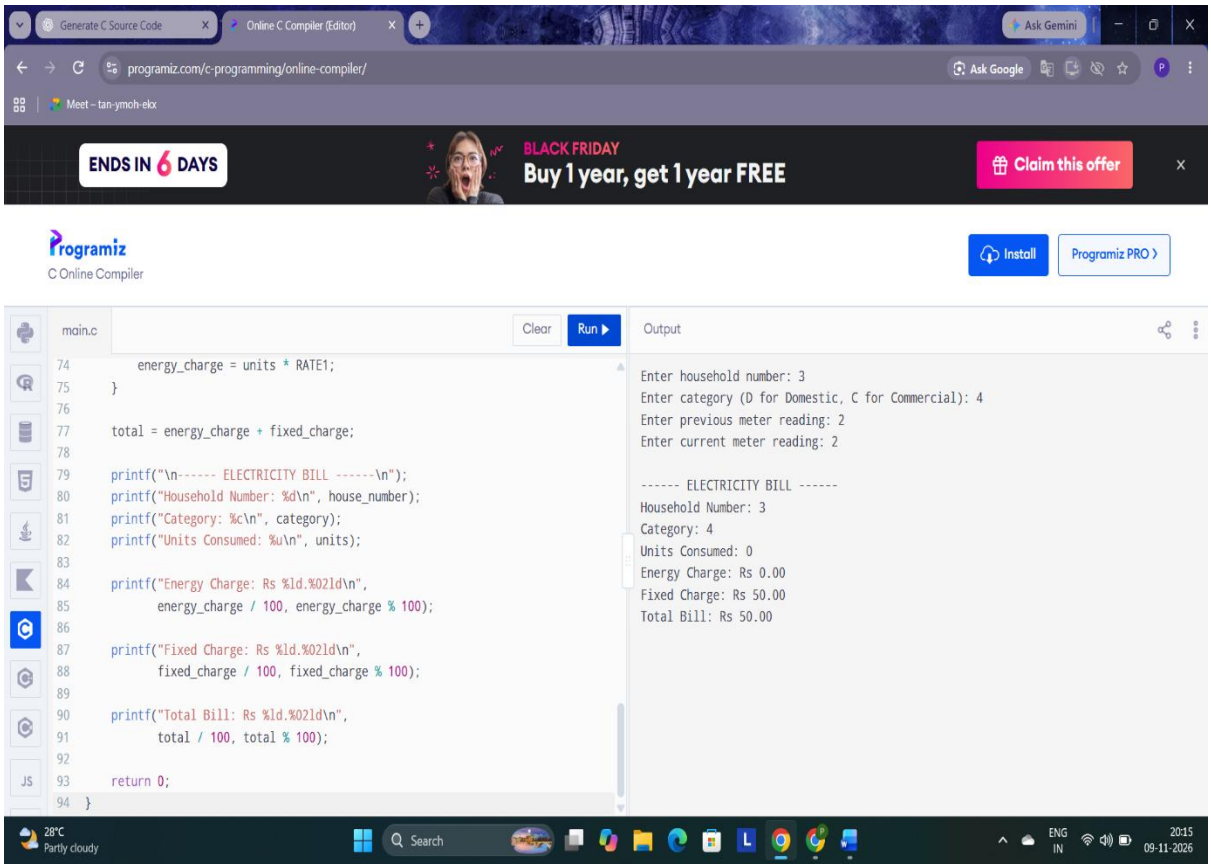
printf("Energy Charge: Rs %ld.%02ld\n",
```

```
        energy_charge / 100, energy_charge % 100);

    printf("Fixed Charge: Rs %ld.%02ld\n",
        fixed_charge / 100, fixed_charge % 100);

    printf("Total Bill: Rs %ld.%02ld\n",
        total / 100, total % 100);

    return 0;
}
```



2) #include <stdio.h>

#include <limits.h>

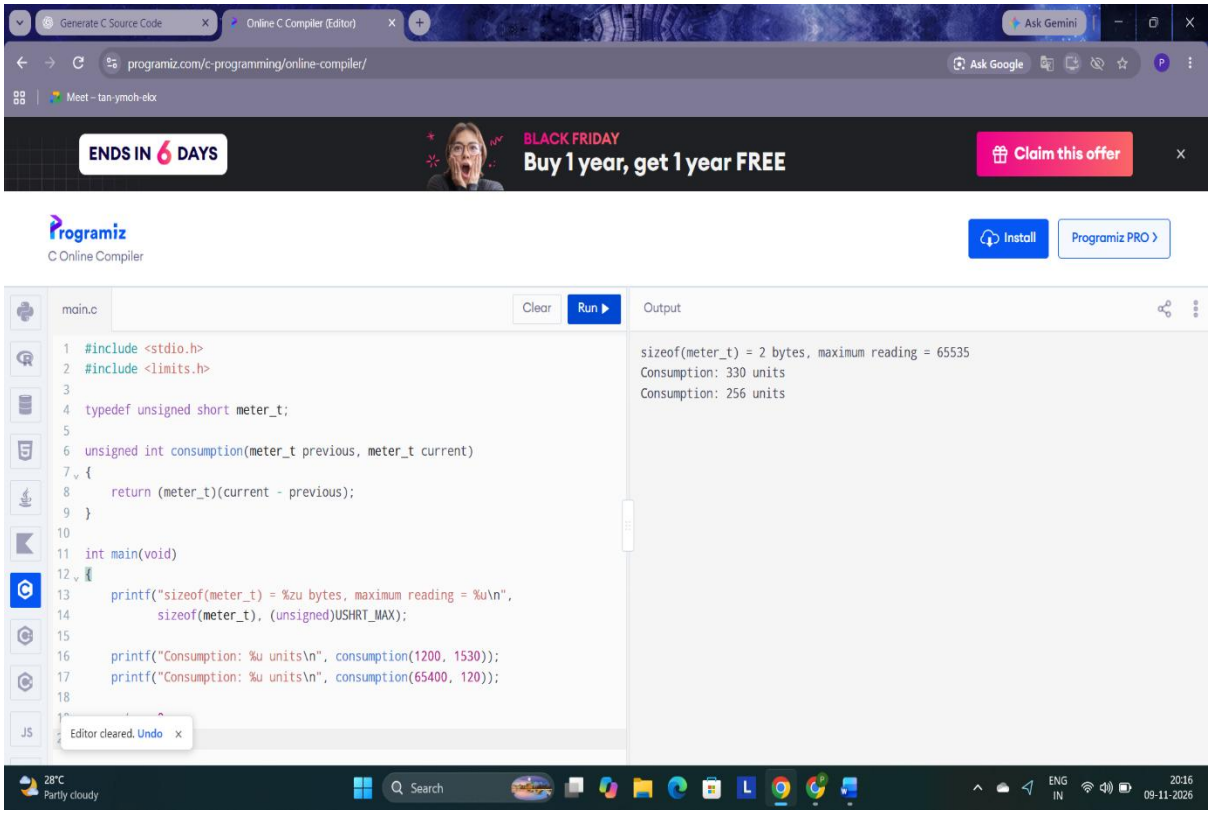
typedef unsigned short meter_t;

```
unsigned int consumption(meter_t previous, meter_t current)
{
   return (meter_t)(current - previous);
}

int main(void)
{
   printf("sizeof(meter_t) = %zu bytes, maximum reading = %u\n",
         sizeof(meter_t), (unsigned)USHRT_MAX);

   printf("Consumption: %u units\n", consumption(1200, 1530));
   printf("Consumption: %u units\n", consumption(65400, 120));

   return 0;
}
```



3) #include <stdio.h>

#define SLAB1_UNITS 100

#define SLAB2_UNITS 200

#define RATE1_PAISE 0

#define RATE2_PAISE 225

#define RATE3_PAISE 450

#define FIXED_CHARGE_PAISE 5000L

long compute_bill(int units)

{

long paise = FIXED_CHARGE_PAISE;

```
    if (units > SLAB2_UNITS)
    {
        paise += (units - SLAB2_UNITS) * RATE3_PAISE;
        paise += (SLAB2_UNITS - SLAB1_UNITS) * RATE2_PAISE;
    }
    else if (units > SLAB1_UNITS)
    {
        paise += (units - SLAB1_UNITS) * RATE2_PAISE;
    }

    return paise;
}

int main(void)
{
    int tests[] = {0, 100, 150, 200, 350};

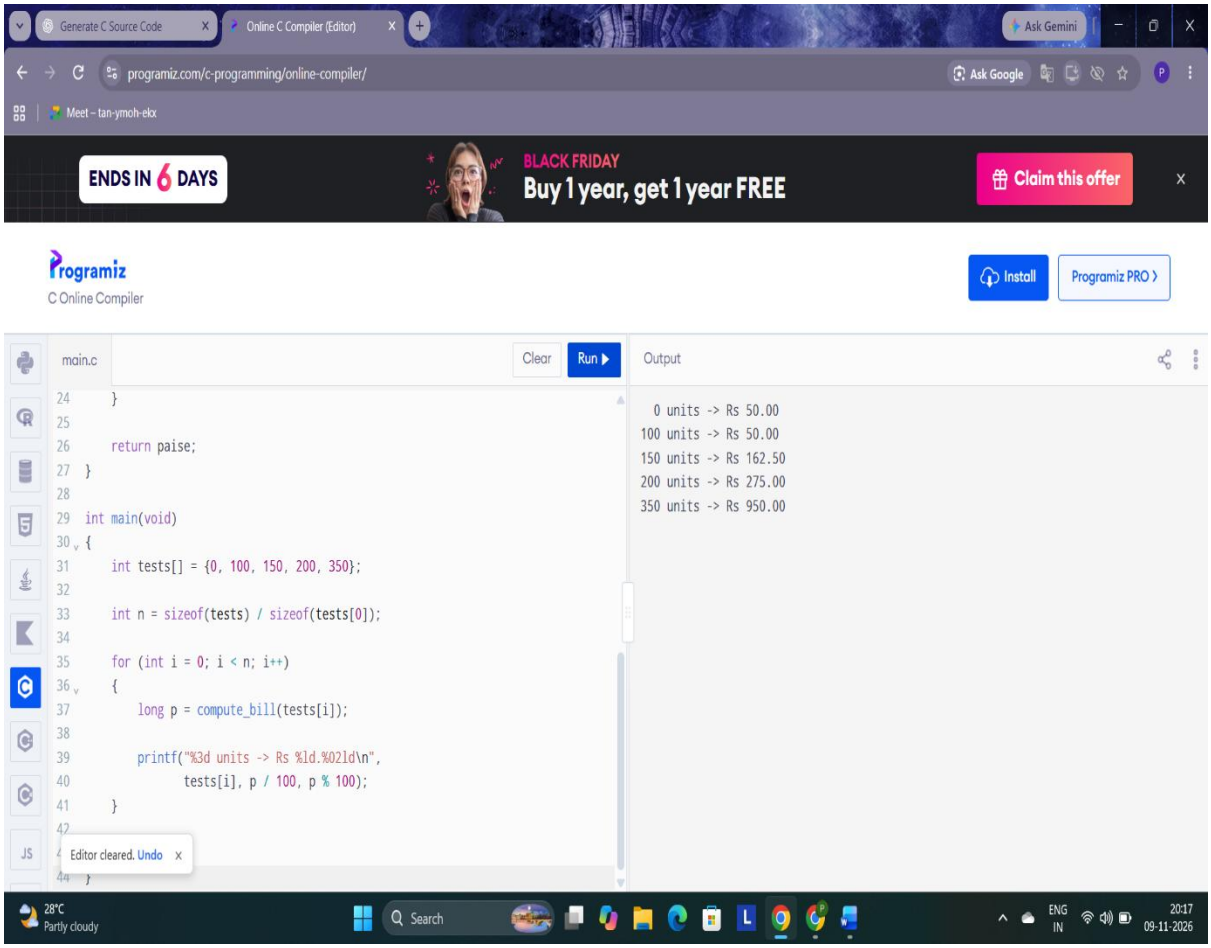
    int n = sizeof(tests) / sizeof(tests[0]);

    for (int i = 0; i < n; i++)
    {
        long p = compute_bill(tests[i]);

        printf("%3d units -> Rs %ld.%02ld\n",
               tests[i], p / 100, p % 100);
    }

    return 0;
```

}



4) #include <stdio.h>

#define SQUARE(x) ((x) * (x))

#define MAX(a, b) ((a) > (b) ? (a) : (b))

#define RUPEES_TO_PAISE(x) ((x) * 100)

static inline int square_function(int x)

{

return x * x;

}

```
static inline int max_function(int a, int b)
{
   return (a > b) ? a : b;
}

int main(void)
{
   int a = 5;
   int b = 10;

   printf("Square of 5 = %d\n", SQUARE(a));
   printf("Maximum value = %d\n", MAX(a, b));

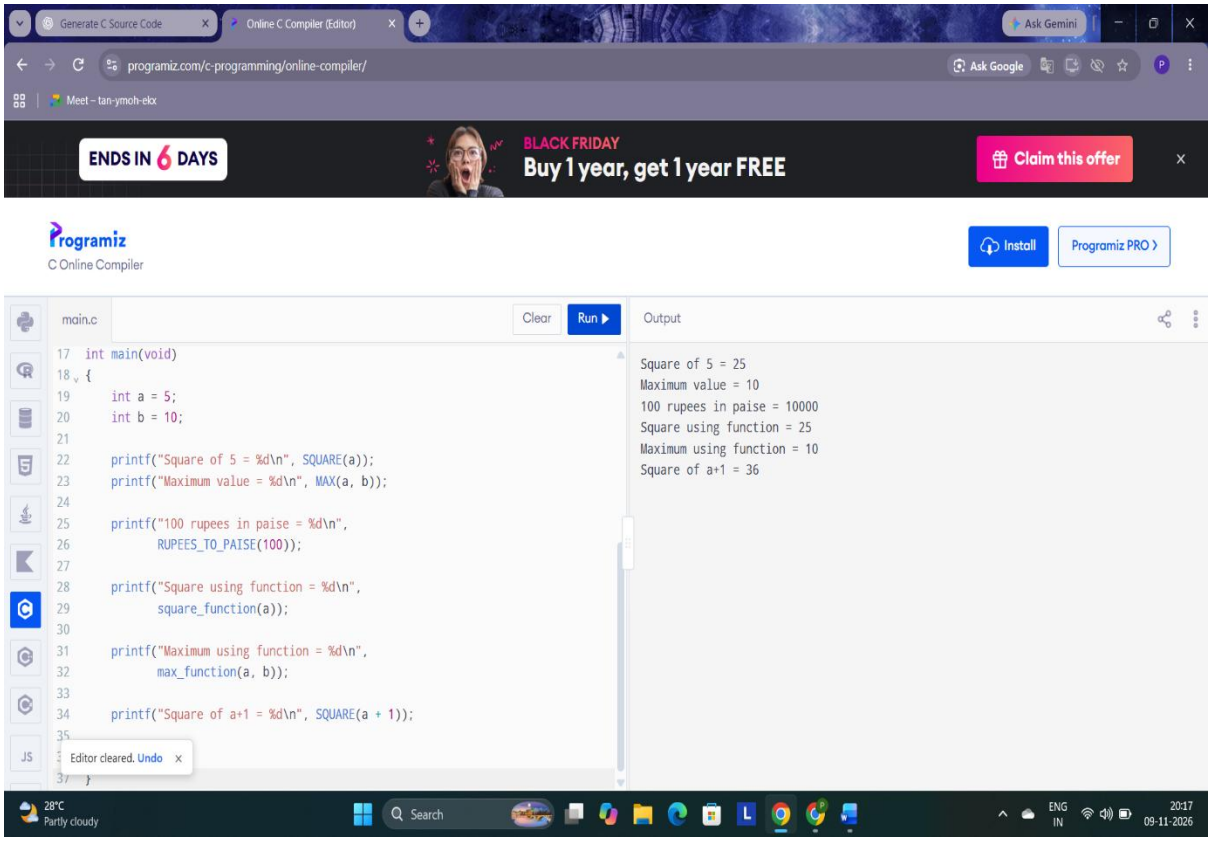
   printf("100 rupees in paise = %d\n",
        RUPEES_TO_PAISE(100));

   printf("Square using function = %d\n",
        square_function(a));

   printf("Maximum using function = %d\n",
        max_function(a, b));

   printf("Square of a+1 = %d\n", SQUARE(a + 1));

   return 0;
}
```



5) #include <stdio.h>

#include "billing.h"

long calculate_bill(int units)

{

long bill = FIXED_CHARGE;

if (units > 200)

{

bill += (units - 200) * RATE3;

bill += 100 * RATE2;

}

else if (units > 100)

{

```
        bill += (units - 100) * RATE2;
    }

#ifdef DEBUG
    printf("DEBUG: Units = %d\n", units);
    printf("DEBUG: Bill in paise = %ld\n", bill);
#endif

    return bill;
}

int main(void)
{
    int units = 150;

    long bill = calculate_bill(units);

    printf("Units consumed = %d\n", units);
    printf("Electricity bill = Rs %ld.%02ld\n",
           bill / 100, bill % 100);

    return 0;
}
```



6) #include <stdio.h>

int main(void)

```
{
    int household;
    unsigned int units;
    long bill_paise;

    printf("Enter household number: ");

    if (scanf("%d", &household) != 1)
    {
        printf("Invalid household number!\n");
        return 1;
    }

    printf("Enter units consumed: ");

    if (scanf("%u", &units) != 1)
    {
        printf("Invalid units!\n");
        return 1;
    }

    printf("Enter bill amount in paise: ");

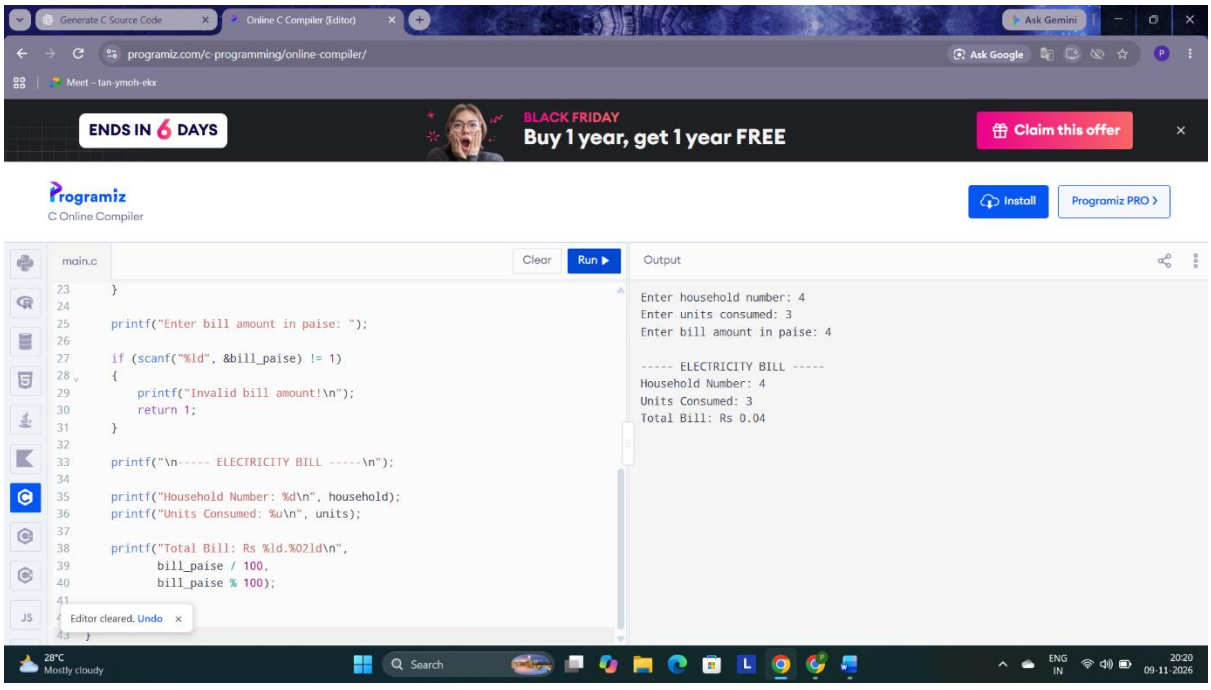
    if (scanf("%ld", &bill_paise) != 1)
    {
        printf("Invalid bill amount!\n");
        return 1;
    }
```

```
    printf("\n----- ELECTRICITY BILL -----\n");

    printf("Household Number: %d\n", household);
    printf("Units Consumed: %u\n", units);

    printf("Total Bill: Rs %ld.%02ld\n",
           bill_paise / 100,
           bill_paise % 100);

    return 0;
}
```



7) #include <stdio.h>

int main(void)

```
{
    int a = 5;
    int b = 2;

    float result1;
    float result2;

    result1 = a / b;
    result2 = (float)a / b;

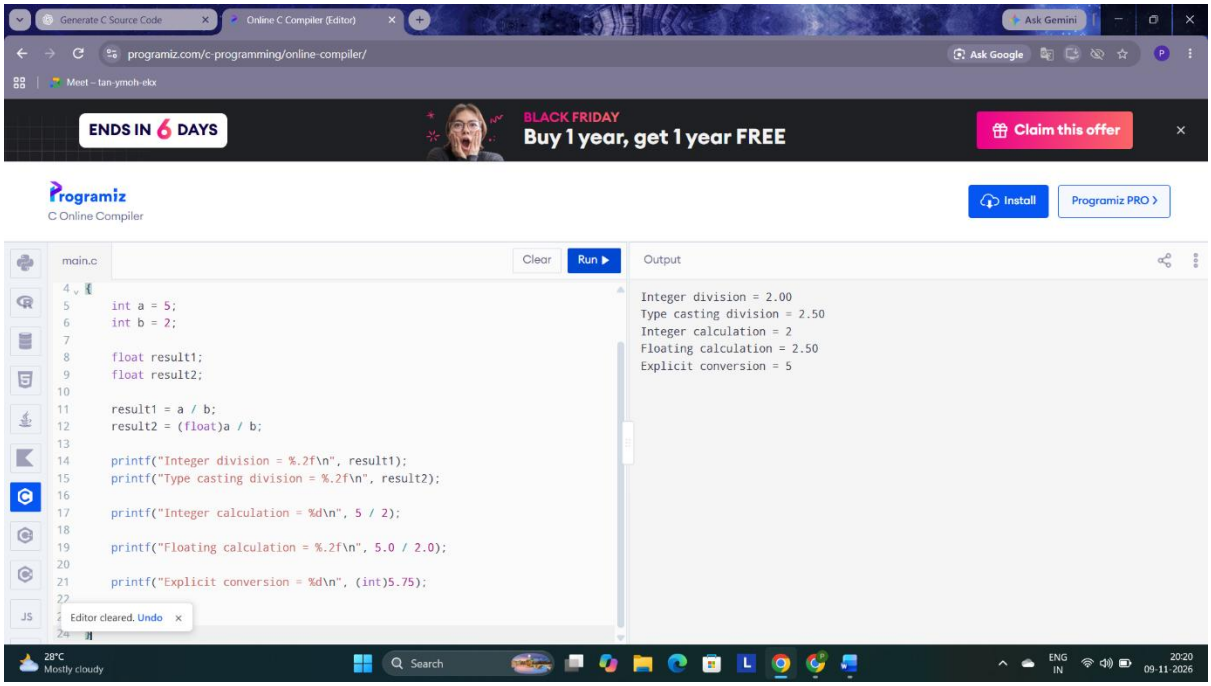
    printf("Integer division = %.2f\n", result1);
    printf("Type casting division = %.2f\n", result2);

    printf("Integer calculation = %d\n", 5 / 2);

    printf("Floating calculation = %.2f\n", 5.0 / 2.0);

    printf("Explicit conversion = %d\n", (int)5.75);

    return 0;
}
```



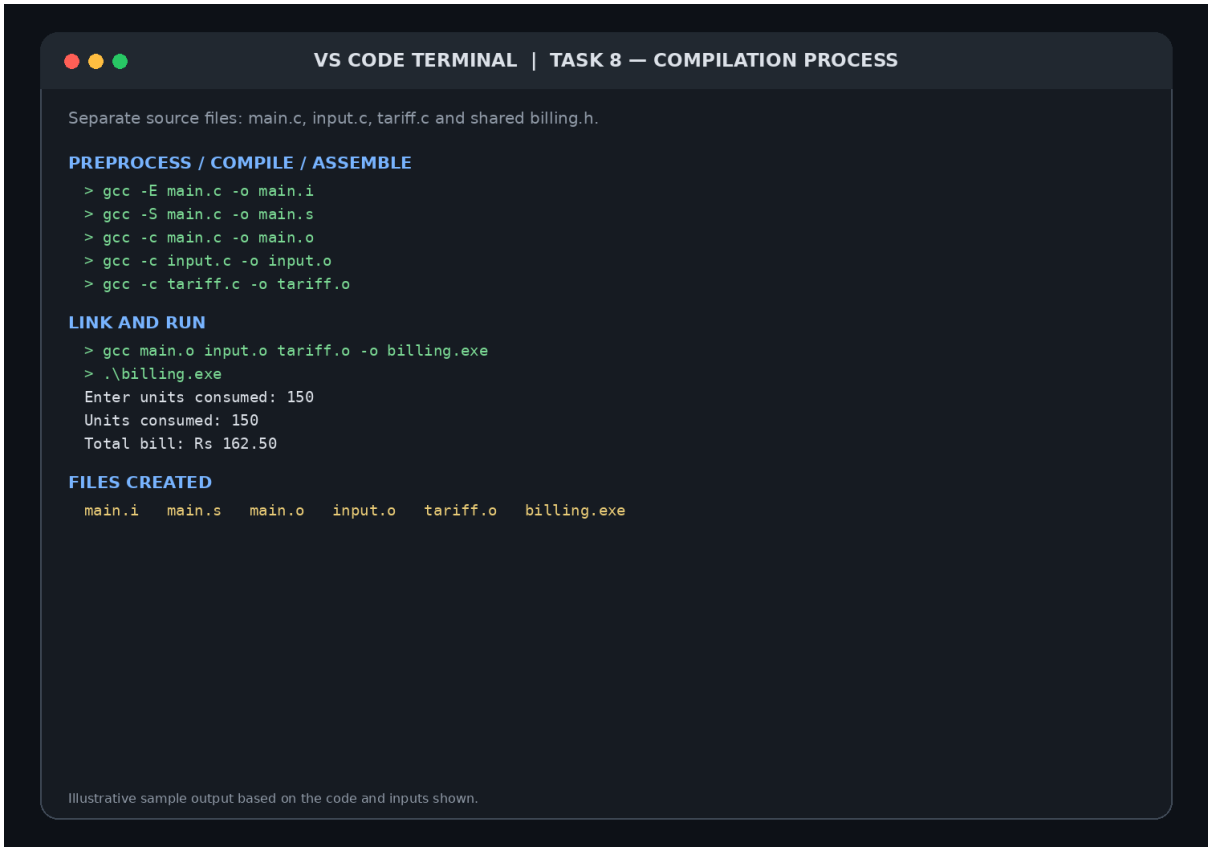
8) #include <stdio.h>

```
int get_units(void)
{
   int units;

   printf("Enter units consumed: ");

   if (scanf("%d", &units) != 1 || units < 0)
   {
      printf("Invalid input\n");
      return -1;
   }

   return units;
}
```



9) #include <stdio.h>

#include <limits.h>

#define MAX_HOUSES 100

#define METER_MAX 65535

#define SLAB1 100

#define SLAB2 200

#define RATE1 0

#define RATE2 225

#define RATE3 450

```
#define FIXED_CHARGE 5000L

typedef unsigned short meter_t;

struct Bill
{
   int household;
   unsigned int units;
   long long amount;
};

unsigned int consumption(meter_t previous, meter_t current)
{
   return (meter_t)(current - previous);
}

long long calculate_bill(unsigned int units)
{
   long long bill = FIXED_CHARGE;

   if (units > SLAB2)
   {
      bill += (units - SLAB2) * RATE3;
      bill += (SLAB2 - SLAB1) * RATE2;
   }
   else if (units > SLAB1)
   {
```

```
        bill += (units - SLAB1) * RATE2;
    }

    return bill;
}

void display_bill(struct Bill b)
{
    printf("\nHousehold Number: %d\n", b.household);
    printf("Units Consumed: %u\n", b.units);

    printf("Total Bill: Rs %lld.%02lld\n",
           b.amount / 100,
           b.amount % 100);
}

int main(void)
{
    struct Bill bills[MAX_HOUSES];

    int n;

    long long total_collection = 0;

    printf("Enter number of households: ");

    if (scanf("%d", &n) != 1 || n < 1 || n > MAX_HOUSES)
    {
```

```
    printf("Invalid number of households!\n");
    return 1;
}

for (int i = 0; i < n; i++)
{
    long previous, current;

    printf("\nHousehold %d\n", i + 1);

    printf("Enter household number: ");

    if (scanf("%d", &bills[i].household) != 1 ||
        bills[i].household <= 0)
    {
        printf("Invalid household number!\n");
        return 1;
    }

    printf("Enter previous reading: ");

    if (scanf("%ld", &previous) != 1 ||
        previous < 0 || previous > METER_MAX)
    {
        printf("Invalid previous reading!\n");
        return 1;
    }
```

```
    printf("Enter current reading: ");

    if (scanf("%ld", &current) != 1 ||
        current < 0 || current > METER_MAX)
    {
        printf("Invalid current reading!\n");
        return 1;
    }

    if (current < previous &&
        !(previous >= 65000 && current <= 535))
    {
        printf("Invalid reading: rollover condition not met!\n");
        return 1;
    }

    bills[i].units = consumption(
        (meter_t)previous,
        (meter_t)current
    );

    bills[i].amount = calculate_bill(bills[i].units);

    total_collection += bills[i].amount;

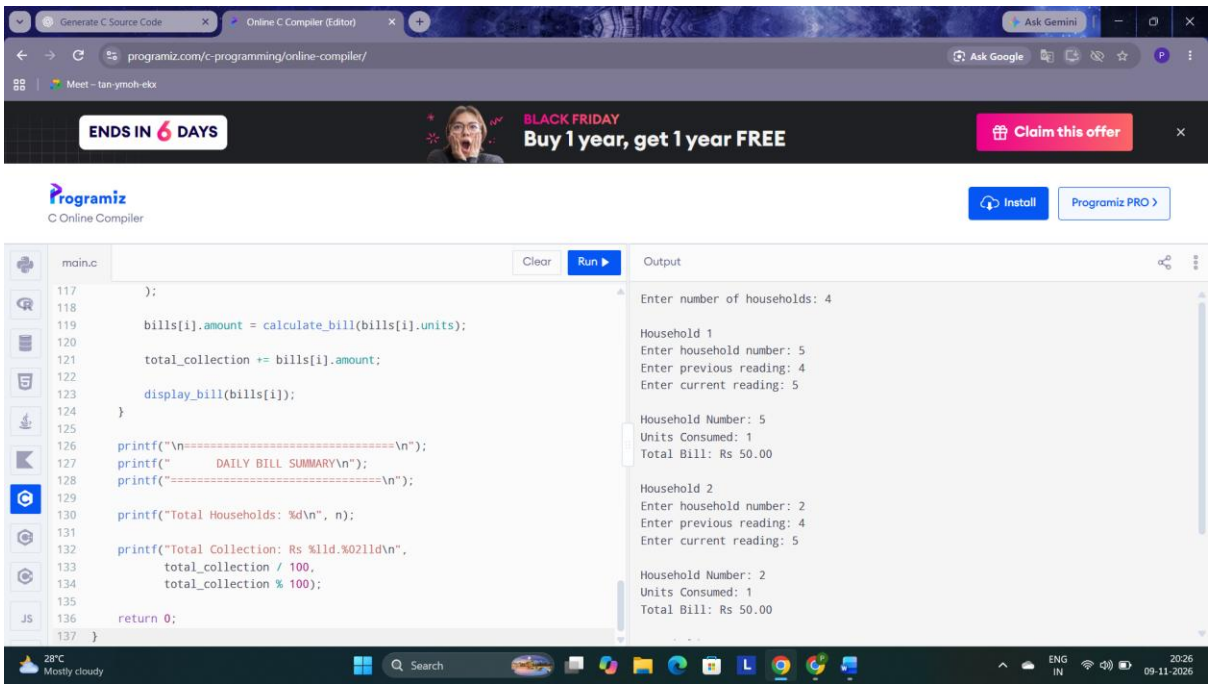
    display_bill(bills[i]);
}
```

```
    printf("\n================================\n");
    printf("      DAILY BILL SUMMARY\n");
    printf("================================\n");

    printf("Total Households: %d\n", n);

    printf("Total Collection: Rs %lld.%02lld\n",
           total_collection / 100,
           total_collection % 100);

    return 0;
}
```



10) #include <stdio.h>

#include <limits.h>

#include <float.h>

int main(void)

```
{
    int household_counts[] = {4000, 60000, 800000};

    long long bill_paise = 95000;

    printf("SIZE OF DATA TYPES\n");
    printf("int      = %zu bytes\n", sizeof(int));
    printf("long     = %zu bytes\n", sizeof(long));
    printf("long long = %zu bytes\n", sizeof(long long));
    printf("float    = %zu bytes\n", sizeof(float));
    printf("double   = %zu bytes\n", sizeof(double));

    printf("\nDATA TYPE LIMITS\n");
    printf("INT_MAX      = %d\n", INT_MAX);
    printf("LONG_MAX     = %ld\n", LONG_MAX);
    printf("LLONG_MAX    = %lld\n", LLONG_MAX);
    printf("FLT_DIG      = %d\n", FLT_DIG);
    printf("DBL_DIG      = %d\n", DBL_DIG);

    printf("\nTOTAL BILL COMPARISON\n");

    for (int i = 0; i < 3; i++)
    {
        int n = household_counts[i];

        long long paise_total = 0;
        float float_total = 0.0f;
```

```
    for (int j = 0; j < n; j++)
    {
        paise_total += bill_paise;
        float_total += 950.0f;
    }

    printf("\nHouseholds: %d\n", n);

    printf("Integer paise total: Rs %lld.%02lld\n",
           paise_total / 100,
           paise_total % 100);

    printf("Float total: Rs %.2f\n", float_total);

    printf("Difference: Rs %.2f\n",
           float_total - (float)paise_total / 100.0f);

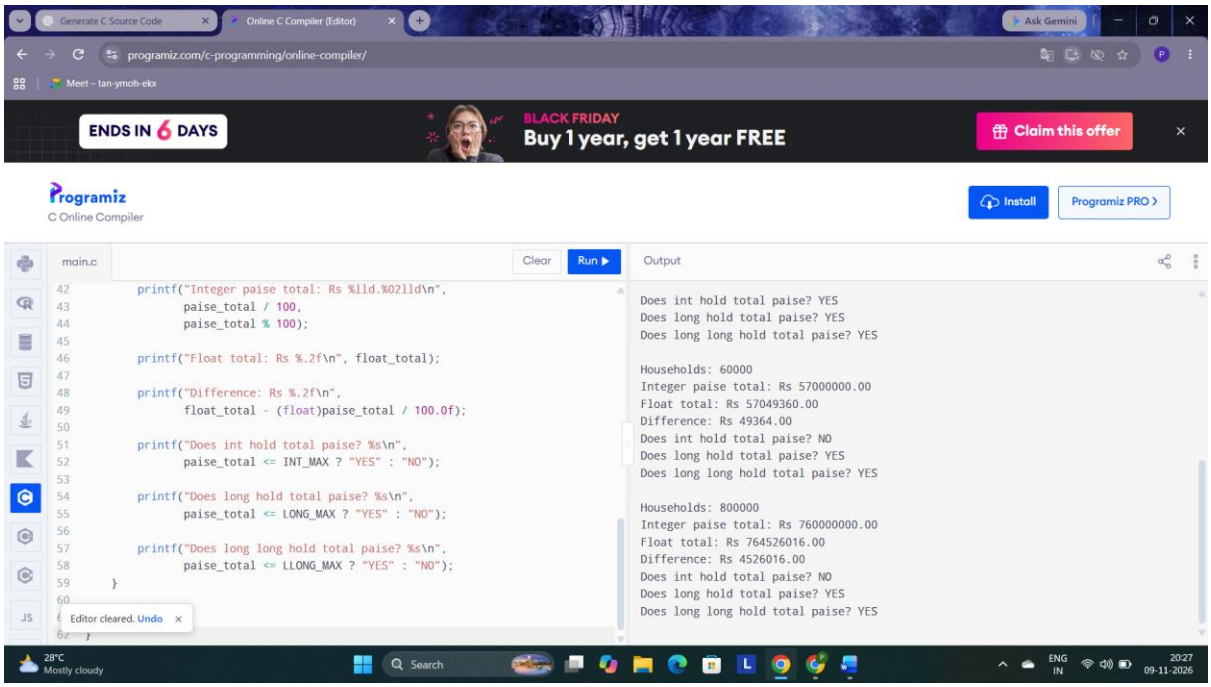
    printf("Does int hold total paise? %s\n",
           paise_total <= INT_MAX ? "YES" : "NO");

    printf("Does long hold total paise? %s\n",
           paise_total <= LONG_MAX ? "YES" : "NO");

    printf("Does long long hold total paise? %s\n",
           paise_total <= LLONG_MAX ? "YES" : "NO");
}

return 0;
```

}



11) #include <stdio.h>

#define MAX_HOUSES 100

#define METER_MAX 65535

typedef unsigned short meter_t;

struct Tariff

{

int slab1;

int slab2;

int rate1;

int rate2;

int rate3;

long fixed_charge;

};

```
long long calculate_bill(unsigned int units,
                         struct Tariff t)
{
    long long bill = t.fixed_charge;

    if (units > t.slab2)
    {
        bill += (long long)(t.slab2 - t.slab1) * t.rate2;
        bill += (long long)(units - t.slab2) * t.rate3;
    }
    else if (units > t.slab1)
    {
        bill += (long long)(units - t.slab1) * t.rate2;
    }
    else
    {
        bill += (long long)units * t.rate1;
    }

    return bill;
}

int main(void)
{
    struct Tariff tariff;

    int n;
```

```
long long total_collection = 0;

FILE *fp = fopen("tariff.txt", "r");

if (fp == NULL)
{
   printf("Cannot open tariff.txt\n");
   return 1;
}

if (fscanf(fp, "%d %d %d %d %d %ld",
        &tariff.slab1,
        &tariff.slab2,
        &tariff.rate1,
        &tariff.rate2,
        &tariff.rate3,
        &tariff.fixed_charge) != 6)
{
   printf("Invalid tariff file\n");
   fclose(fp);
   return 1;
}

fclose(fp);

if (tariff.slab1 < 0 ||
   tariff.slab2 <= tariff.slab1 ||
   tariff.rate1 < 0 ||
```

```
    tariff.rate2 < 0 ||
    tariff.rate3 < 0 ||
    tariff.fixed_charge < 0)
{
    printf("Invalid tariff values\n");
    return 1;
}

printf("Enter number of households: ");

if (scanf("%d", &n) != 1 ||
    n < 1 || n > MAX_HOUSES)
{
    printf("Invalid number of households\n");
    return 1;
}

for (int i = 0; i < n; i++)
{
    long previous, current;
    unsigned int units;
    long long bill;

    printf("\nHousehold %d\n", i + 1);

    printf("Enter previous reading: ");

    if (scanf("%ld", &previous) != 1 ||
```

```
   previous < 0 || previous > METER_MAX)
{
   printf("Invalid previous reading\n");
   return 1;
}

printf("Enter current reading: ");

if (scanf("%ld", &current) != 1 ||
   current < 0 || current > METER_MAX)
{
   printf("Invalid current reading\n");
   return 1;
}

if (current < previous &&
   !(previous >= 65000 && current <= 535))
{
   printf("Invalid meter rollover\n");
   return 1;
}

units = (meter_t)(
   (meter_t)current - (meter_t)previous
);

bill = calculate_bill(units, tariff);
```

```
    total_collection += bill;

    printf("Units consumed: %u\n", units);

    printf("Bill amount: Rs %lld.%02lld\n",
           bill / 100, bill % 100);
}

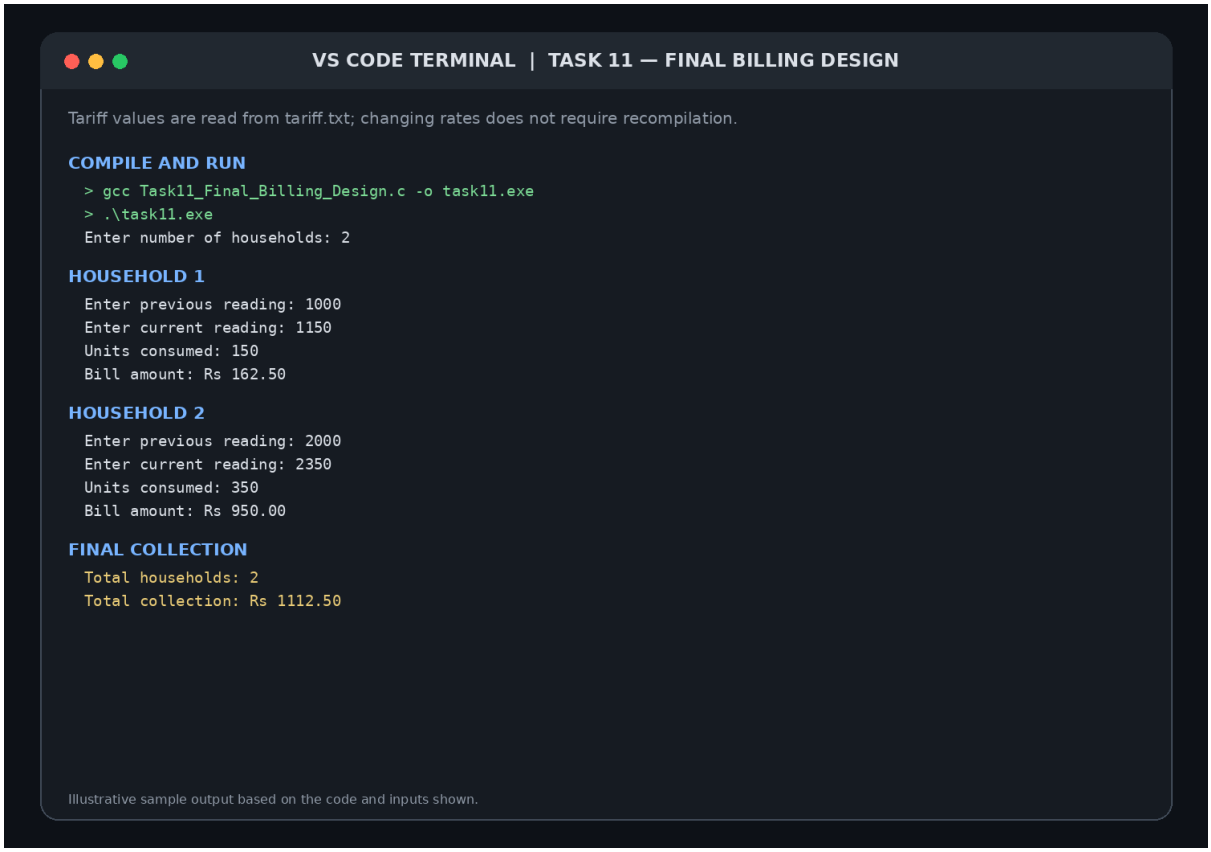
printf("\n----- FINAL COLLECTION -----\n");

printf("Total households: %d\n", n);

printf("Total collection: Rs %lld.%02lld\n",
       total_collection / 100,
       total_collection % 100);

return 0;
```

}



12) #include <stdio.h>

int main(void)

{

double normal_rate = 2.25;

double peak_rate;

double normal_energy;

double peak_energy;

double total_bill;

```
    printf("Enter normal energy in Wh: ");
    scanf("%lf", &normal_energy);

    printf("Enter peak energy in Wh: ");
    scanf("%lf", &peak_energy);

    peak_rate = normal_rate * 1.20;

    normal_energy = normal_energy / 1000.0;
    peak_energy = peak_energy / 1000.0;

    total_bill = (normal_energy * normal_rate)
            + (peak_energy * peak_rate);

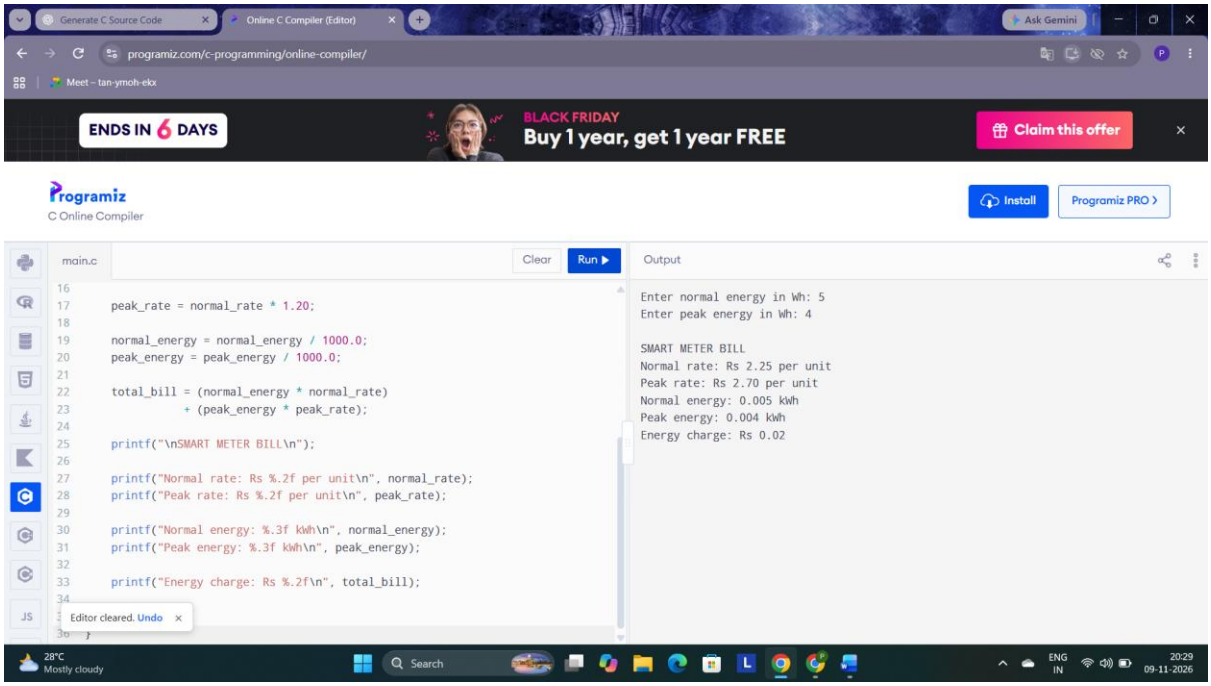
    printf("\nSMART METER BILL\n");

    printf("Normal rate: Rs %.2f per unit\n", normal_rate);
    printf("Peak rate: Rs %.2f per unit\n", peak_rate);

    printf("Normal energy: %.3f kWh\n", normal_energy);
    printf("Peak energy: %.3f kWh\n", peak_energy);

    printf("Energy charge: Rs %.2f\n", total_bill);

    return 0;
}
```


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C Online Compiler
Install
Programiz PRO
main.c
Clear
Run
Output
peak_rate = normal_rate * 1.20;
normal_energy = normal_energy / 1000.0;
peak_energy = peak_energy / 1000.0;
total_bill = (normal_energy * normal_rate)
+ (peak_energy * peak_rate);
printf("\nSMART METER BILL\n");
printf("Normal rate: Rs %.2f per unit\n", normal_rate);
printf("Peak rate: Rs %.2f per unit\n", peak_rate);
printf("Normal energy: %.3f kWh\n", normal_energy);
printf("Peak energy: %.3f kWh\n", peak_energy);
printf("Energy charge: Rs %.2f\n", total_bill);
Editor cleared. Undo
Enter normal energy in Wh: 5
Enter peak energy in Wh: 4
SMART METER BILL
Normal rate: Rs 2.25 per unit
Peak rate: Rs 2.70 per unit
Normal energy: 0.005 kWh
Peak energy: 0.004 kWh
Energy charge: Rs 0.02
28°C
Mostly cloudy
Search
ENG
IN
20:29
09-11-2026